

PSYCHOLOGY 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: What a Baby Already Knows

Grading guide. Total 22 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Design an experiment

1. You want to find out whether a newborn recognizes her father's voice. Describe an experiment: what would you play for the baby, and what would you measure to tell whether she recognizes it? (3 pts)

Answer: Play the father's voice and a stranger's voice (1 pt for comparing two voices); measure something the baby does — sucking pattern, heart rate change, head turning, or looking time (1 pt); if the baby responds differently to dad's voice (e.g., sucks to hear more of it, calms, turns), she recognizes it (1 pt).

Part B — Match the method to what it measures

2. Draw a line from each baby-science method to what it tells us. (4 pts)

Answer: Looking time -> Whether a baby finds something new or interesting | Sucking rate -> Which of two sounds a baby prefers to hear | Heart rate change -> Whether a baby is startled or calmed by something | Head turning -> Which direction a sound or smell is coming from (1 pt each)

Part C — True or false?

3. Write TRUE or FALSE for each statement about newborns. (4 pts)

Answer: TRUE (hearing works from about 24-25 weeks); FALSE (newborns focus best at about 20-30 cm — a parent's face); TRUE; FALSE (newborns clearly feel pain — measured by stress responses — and NICUs treat it).

Part D — Invent a score (Statistics flashback)

4. Dr. Apgar turned 'how is this baby?' into a number. Invent a tiny COMFORT score for a NICU baby using two signs you could see. Fill in what 0, 1, and 2 points would look like for each sign. (2 pts)

Answer: Any two observable signs with sensible 0/1/2 levels — e.g., Face: crying/grimacing (0), frowning (1), relaxed (2); Body: stiff and squirming (0), some wiggling (1), relaxed and still (2); or breathing, color, sleep. 1 pt per sensible row.

Part E — Faces

5. Draw two simple ovals. In the first, put two dots and a mouth-line where a face's eyes and mouth go. In the second, put the same dots and line in scrambled places. Circle the one a newborn would look at longer. (2 pts)

Answer: Two ovals, one face-like and one scrambled (1 pt); the face-like one circled (1 pt). Newborns prefer face-like patterns from the first days.

6. Why is kangaroo care called brain care? (1 pt)

Answer: A. Calm skin-to-skin contact steadies stress, sleep, and heart rate while the brain is wiring fast. - A preemie does its third-trimester brain wiring in the NICU; calm and sleep are what that wiring needs.

Part F — Flashback

7. From NICU Tutor: name TWO things kangaroo care does for a baby. (2 pts)

Answer: Any two: keeps the baby warm, steadies heart rate, steadies breathing, helps with feeding/milk supply, helps bonding, calms the baby.

8. From Genetics: identical twins share exactly the same _____. (1 pt)

Answer: DNA (instruction book).

Part G — Think like a doctor

9. You are designing a new NICU. Name THREE things you would change about a typical hospital room to make it gentler for preemies' developing brains, and say why each helps. (3 pts)

Answer: 1 pt per sensible change with a reason, e.g.: dim/adjustable lights (preemies' eyes and sleep); quieter alarms and soft voices (they hear everything from 24 weeks); cluster care so babies aren't woken repeatedly (protect sleep); room and chairs for parents to do kangaroo care (calm, heartbeat, voice); covers over isolettes; private rooms.